DEEP LEARNING

Quiz Question Bank
Chapter-3,4,5

Linear Regression • Gradient Descent • Bias & MSE • Activation Functions • Forward & Backpropagation

Notation guide (important for forward/backward propagation)

- Hidden layer: $z_1 = w_1x + b_1$, $h_1 = \text{ReLU}(z_1)$; $z_2 = w_2x + b_2$, $h_2 = \text{ReLU}(z_2)$. These are the simple one-input-to-one-hidden-unit connections used for hand calculation.
- Output layer for regression: $\hat{y} = v_1h_1 + v_2h_2 + c$. In the supplied notes, the output weights are named v_1 and v_2, and c is the output bias.
- If your class calls the two outgoing weights c_1 and c_2, use $c_1 \equiv v_1$ and $c_2 \equiv v_2$. Keep c as the output bias; b_1 and b_2 are hidden-layer biases.
- For one sample, half-squared loss is $L = \frac{1}{2}(\hat{y} - y)^2$ and $\partial L/\partial \hat{y} = \hat{y} - y$. Dataset MSE is $\text{MSE} = (1/m)\Sigma(\hat{y}_i - y_i)^2$; these formulas differ by the factor ½ and by averaging over samples.
- The gradient is calculated by backpropagation. Gradient descent then updates a parameter: $\theta_new = \theta_old - \eta(\partial L/\partial \theta)$. Do not confuse calculating gradients with updating parameters.

Core formulas to revise

Concept	Formula
Linear regression prediction	$\hat{y} = w_1x_1 + w_2x_2 + b = w^Tx + b$
Residual	$r = y - \hat{y}$
Mean squared error	$\text{MSE} = (1/m)\ \Sigma_i\ (\hat{y}_i - y_i)^2$
Gradient descent	$\theta_new = \theta_old - \eta\nabla\theta L$
Gradient descent for $f(x)=x^2$	$f'(x)=2x;\ x_new = x_old - \eta(2x_old)$
ReLU	$\text{ReLU}(z) = \max(0,z)$
Sigmoid	$\sigma(z) = 1/(1+e^{\wedge}(-z))$
Half-squared loss	$L = \frac{1}{2}(\hat{y} - y)^2;\ \partial L/\partial \hat{y} = \hat{y} - y$
Linear output with two hidden units	$\hat{y} = v_1h_1 + v_2h_2 + c$
Output-layer gradients	$\partial L/\partial v_1 = (\hat{y}-y)h_1;\ \partial L/\partial v_2 = (\hat{y}-y)h_2;\ \partial L/\partial c = \hat{y}-y$
Hidden-layer gradient (active ReLU)	$\partial L/\partial h_i = (\hat{y}-y)v_i;\ \partial L/\partial w_i = (\partial L/\partial h_i)\text{ReLU}'(z_i)x$
Binary cross-entropy	$L = -[y \ln p + (1-y)\ln(1-p)]$
Softmax	$p_i = e^{\wedge}(z_i) / \Sigma_j\ e^{\wedge}(z_j)$

Part I - short conceptual questions with answers

1. What is the difference between regression and classification?

Answer: Regression predicts a continuous numerical value, such as training time. Classification predicts a discrete class label, such as benign versus malicious.

2. What is a feature, and what is a target?

Answer: A feature is an input attribute supplied to the model; the target is the true value the model is intended to predict. For the training-time example, epochs and batch-size units are features, while observed training minutes are the target.

3. What does a weight mean in linear regression?

Answer: A weight describes the predicted change in output for a one-unit change in its feature, holding other features fixed. In $\hat{y} = 2x_1 + 3x_2 + 5$, the model adds 2 predicted minutes per extra epoch and 3 per batch-size unit.

4. Why include a bias term?

Answer: The bias shifts the line or plane independently of the input. Without bias, the linear model $\hat{y}=wx$ is forced through the origin; with bias, it can have a nonzero intercept.

5. What is a feature vector?

Answer: It packages all features for one observation into a vector, such as $x=[x_1,x_2]^T$. It provides a compact way to write the prediction as $\hat{y}=w^Tx+b$.

6. What are model parameters?

Answer: They are learnable values adjusted during training. In these simple models, the parameters include the weights and bias, written $\theta=\{w_1,w_2,\ldots,b\}$.

7. What is a residual? Why does its sign matter?

Answer: The residual is $r=y-\hat{y}$, the actual value minus the prediction. A positive residual means the model predicted too low; a negative residual means it predicted too high.

8. Why does MSE square the residuals?

Answer: Squaring prevents positive and negative errors from cancelling and makes large deviations count more heavily. MSE is nonnegative and equals zero when every prediction matches its target.

9. What does gradient descent do?

Answer: It updates parameters in the direction opposite the loss gradient: $\theta_new=\theta_old-\eta\nabla\theta L$. The learning rate η controls the step size.

10. What does the derivative tell us in gradient descent?

Answer: The derivative gives the local slope of the objective. For $f(x)=x^2$, $f'(x)=2x$; at positive x, the slope is positive, so subtracting a positive gradient moves x toward zero.

11. What is the role of the learning rate?

Answer: It scales each gradient step. In the provided $f(x)=x^2$ example, starting at $x_0=4$ with $\eta=0.1$ gives $x_1=4-0.1(8)=3.2$.

12. How does ReLU behave?

Answer: $ReLU(z)=\max(0,z)$. It outputs zero for a negative input and passes a positive input through unchanged; examples include $ReLU(-3)=0$ and $ReLU(2.5)=2.5$.

13. What is a potential limitation of ReLU during backpropagation?

Answer: For a negative pre-activation, the ReLU derivative is zero. The gradient through that unit is blocked for that sample, so its incoming parameters may receive no update through that path.

14. What does sigmoid do, and where is it useful?

Answer: Sigmoid maps a real-valued logit to a probability strictly between 0 and 1. The notes use it for binary classification, where one value represents the probability of class 1.

15. Why is sigmoid(0)=0.5?

Answer: Substituting $z=0$ gives $1/(1+e^0)=1/(1+1)=0.5$. A zero logit therefore gives equal probability to the two binary outcomes.

16. What is the difference between hidden-layer and output-layer parameters?

Answer: In the simple two-hidden-unit example, w_1, w_2, b_1, b_2 control the feature transformations $x \to h$. Output weights v_1, v_2 and output bias c combine h_1, h_2 into $\hat{y}$; these are independent trainable parameters.

17. What does forward propagation calculate?

Answer: It moves from inputs through the hidden layer to the output, calculating pre-activations, activations, prediction and loss. It produces the values needed to start the backward calculation.

18. What does backpropagation calculate?

Answer: It applies the chain rule from the loss backward through the output and hidden layers to calculate each parameter's gradient. An optimizer such as gradient descent uses those gradients to update the parameters.

19. Why is the chain rule essential to backpropagation?

Answer: A parameter affects the loss through intermediate computations. The chain rule multiplies local derivatives along that path, allowing the loss derivative to be computed efficiently.

Part II - 25 True/False questions with reasons

For practice, decide True or False before reading the answer and explanation.

1. Linear regression normally predicts a continuous numerical target.

True. For example: training time, housing prices and temperature.

2. In supervised learning, the target is the input attribute passed to the model.

False. The feature is the input attribute; the target is the true value to predict.

3. A larger weight always means its feature is more important.

False. Weights depend on feature scale; normalization can be needed before comparing magnitudes.

4. The model $\hat{y}=wx$ with no bias must pass through (0,0).

True. At $x=0$, the prediction is zero.

5. In the simple linear model, weights and bias are learnable parameters.

True. Training adjusts these values to reduce the objective.

6. Squaring residuals allows positive and negative errors to cancel.

False. Squaring makes each contribution nonnegative and prevents sign cancellation.

7. MSE is zero when every prediction equals its target.

True. Every squared residual is zero, so their mean is zero.

8. Gradient descent updates parameters by adding the gradient.

False. It subtracts the gradient: $\theta_{new}=\theta_{old}-\eta\nabla\theta L$.

9. For $f(x)=x^2$, the derivative is $f'(x)=2x$.

True. Differentiating x^2 gives $2x$.

10. With $x_0=4$ and $\eta=0.1$ for $f(x)=x^2$, the first gradient-descent value is 3.2.

True. The gradient is 8, so $x_1=4-0.1\times8=3.2$.

11. ReLU(−3)=−3.

False. ReLU returns $\max(0,z)$, so $\text{ReLU}(-3)=0$.

12. For a positive input, ReLU passes the value through unchanged.

True. When $z>0$, $\text{ReLU}(z)=z$ and its derivative is 1.

13. For $z<0$, the ReLU derivative is zero.

True. The inactive negative branch has zero derivative.

14. The sigmoid function outputs exactly 0 or 1 for finite logits.

False. For finite z, sigmoid lies strictly between 0 and 1.

15. Sigmoid(0)=0.5.

True. $\sigma(0)=1/(1+e^0)=1/2$.

16. Softmax probabilities across classes should sum to 1.

True. Softmax normalizes the exponentiated logits into a probability distribution.

17. Softmax is used for mutually exclusive multiclass classification.

True. The example distinguishes one class among several alternatives.

18. The output weights v_1 and v_2 are calculated from hidden weights w_1 and w_2.

False. The output weights are separate learnable parameters; they are initialized and trained independently.

19. Backpropagation and gradient descent are two names for the same operation.

False. Backpropagation calculates gradients; gradient descent uses them to update parameters.

Part III - mathematical problems with worked solutions

Problem 1. Linear regression prediction

Given / Find: $\hat{y} = 2x_1 + 3x_2 + 5$; $x_1=2$, $x_2=2$. Find $\hat{y}$.

Solution:

$\hat{y} = 2(2)+3(2)+5$

$= 4+6+5$

$= 15$.

Why it matters: The prediction is a numerical output; no activation is applied in this linear model.

Problem 2. Residual and squared error

Given / Find: Use the model in Problem 1. If the actual target is y=17, find $r=y-\hat{y}$ and r^2.

Solution:

$\hat{y}=15$

$r=y-\hat{y}=17-15=2$

Squared error = $r^2 = 2^2 = 4$.

Why it matters: A positive residual means the prediction was below the actual target.

Problem 3. First gradient-descent step

Given / Find: Minimize $f(x)=x^2$ from $x_0=4$ with learning rate $\eta=0.1$. Find x_1 and $f(x_1)$.

Solution:

$f'(x)=2x$, so $f'(4)=8$.

$x_1=x_0-\eta f'(x_0)=4-0.1(8)=3.2$.

$f(x_1)=3.2^2=10.24$.

Why it matters: The positive gradient causes x to move left toward the minimum at zero.

Problem 4. Calculate ReLU outputs

Given / Find: Calculate ReLU(−3), ReLU(0), and ReLU(2.5).

Solution:

$\text{ReLU}(z)=\max(0,z)$.

ReLU(−3)=0; ReLU(0)=0; ReLU(2.5)=2.5.

Why it matters: The negative branch is zero; positive values pass through unchanged.

Problem 5. Easy forward pass: hidden units and output

Given / Find: Use $x=1$; $w_1=1, b_1=0$; $w_2=1, b_2=0$; $h_i=\text{ReLU}(w_i x+b_i)$. Output weights $c_1=0.2, c_2=0.2$ and output bias $c=0$. Find h_1, h_2 and $\hat{y}$.

Solution:

$z_1=1(1)+0=1$, so $h_1=\text{ReLU}(1)=1$.

$z_2=1(1)+0=1$, so $h_2=\text{ReLU}(1)=1$.

$\hat{y}=c_1h_1+c_2h_2+c=0.2(1)+0.2(1)+0=0.4$.

Why it matters: The source notes call c_1 and c_2 output weights v_1 and v_2; here $c_1 \equiv v_1$ and $c_2 \equiv v_2$ to match the requested notation.

Problem 6. Half-squared loss from the forward pass

Given / Find: For $\hat{y}=0.4$ and target $y=1$, calculate $L=\frac{1}{2}(\hat{y}-y)^2$.

Solution:

$L=\frac{1}{2}(0.4-1)^2=\frac{1}{2}(-0.6)^2$.

$L=\frac{1}{2}(0.36)=0.18$.

Why it matters: The half factor makes the derivative of this single-sample loss equal to $\hat{y}-y$.

Problem 7. Inactive ReLU

Given / Find: Given $x_1=3, x_2=-1, w_1=1, w_2=2, b_1=b_2=1$. Use $h_1=\text{ReLU}(w_1x_1+b_1)$, $h_2=\text{ReLU}(w_2x_2+b_2)$. If output weights are $v_1=v_2=0.2$ and $c=0$, find h_1, h_2 and $\hat{y}$.

Solution:

$z_1=1(3)+1=4$, so $h_1=4$.

$z_2=2(-1)+1=-1$, so $h_2=0$.

$\hat{y}=0.2(4)+0.2(0)+0=0.8$.

Why it matters: The second hidden unit is inactive for this input, so its outgoing-weight gradient is zero for this example.

Problem 8. Sigmoid probability

Given / Find: Calculate $\sigma(0)$ and approximately calculate $\sigma(2)$, using $\sigma(z)=1/(1+e^{(-z)})$.

Solution:

$\sigma(0)=1/(1+1)=0.5$.

$\sigma(2)=1/(1+e^{(-2)})\approx 1/(1+0.1353)\approx 0.881$.

Why it matters: Sigmoid converts a logit into a number between 0 and 1.

Problem 9. Binary cross-entropy

Given / Find: For predicted class-1 probability $p=0.8$, calculate BCE for (a) true label $y=1$ and (b) true label $y=0$. Use natural logs.

Solution:

For $y=1$: $L=-\ln(0.8)\approx 0.2231$.

For $y=0$: $L=-\ln(1-0.8)=-\ln(0.2)\approx 1.6094$.

Why it matters: The same probability can have very different loss depending on which label is correct.

Problem 10. Softmax probability vector

Given / Find: For logits z=[2,1,0], use $e^2 \approx 7.389$, $e^1 \approx 2.718$ and $e^0 = 1$. Calculate the softmax probabilities.

Solution:

Sum=7.389+2.718+1=11.107.

$p_1 = 7.389/11.107 \approx 0.665$.

$p_2 = 2.718/11.107 \approx 0.245$; $p_3 = 1/11.107 \approx 0.090$.

Check: 0.665+0.245+0.090≈1.000.

Why it matters: Softmax forms a normalized probability distribution across mutually exclusive classes.

Problem 11. Categorical cross-entropy for class 2

Given / Find: Use the softmax result [0.665,0.245,0.090] and true one-hot label [0,1,0]. Calculate the categorical cross-entropy.

Solution:

$L = -\Sigma\, y_k \ln p_k$.

Only the second target entry is 1, so $L = -\ln(0.245)$.

L≈1.407.

Why it matters: The loss depends on the probability assigned to the true class, not just which class has the largest probability.

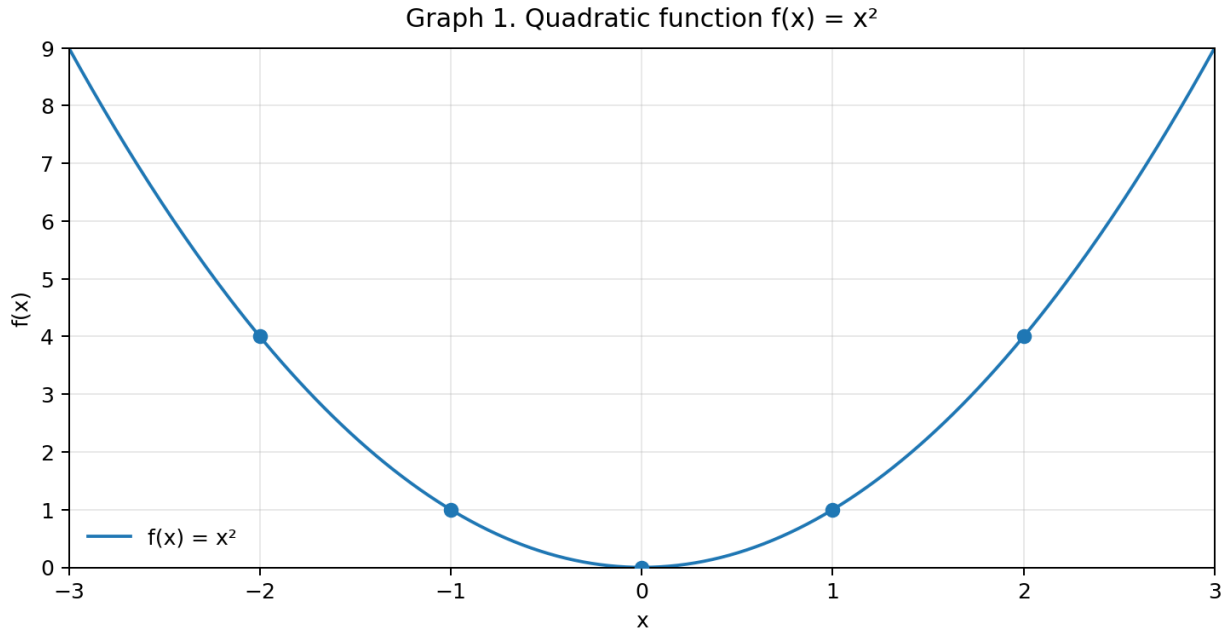
Part IV - 10 graph-plotting exercises with reference graphs

For each exercise, practise drawing axes and labels, plotting the key points, shaping the curve or line correctly, and explaining what the graph shows. Reference plots are provided after the plotting instructions for self-check.

Graph Plotting Exercise 1. Quadratic function $f(x)=x^2$

Task: Plot $f(x)=x^2$ for x from −2 to 2. Label the minimum.

Suggested points / plot instructions: Use points (−2,4), (−1,1), (0,0), (1,1), (2,4). Join smoothly with a U-shaped curve.

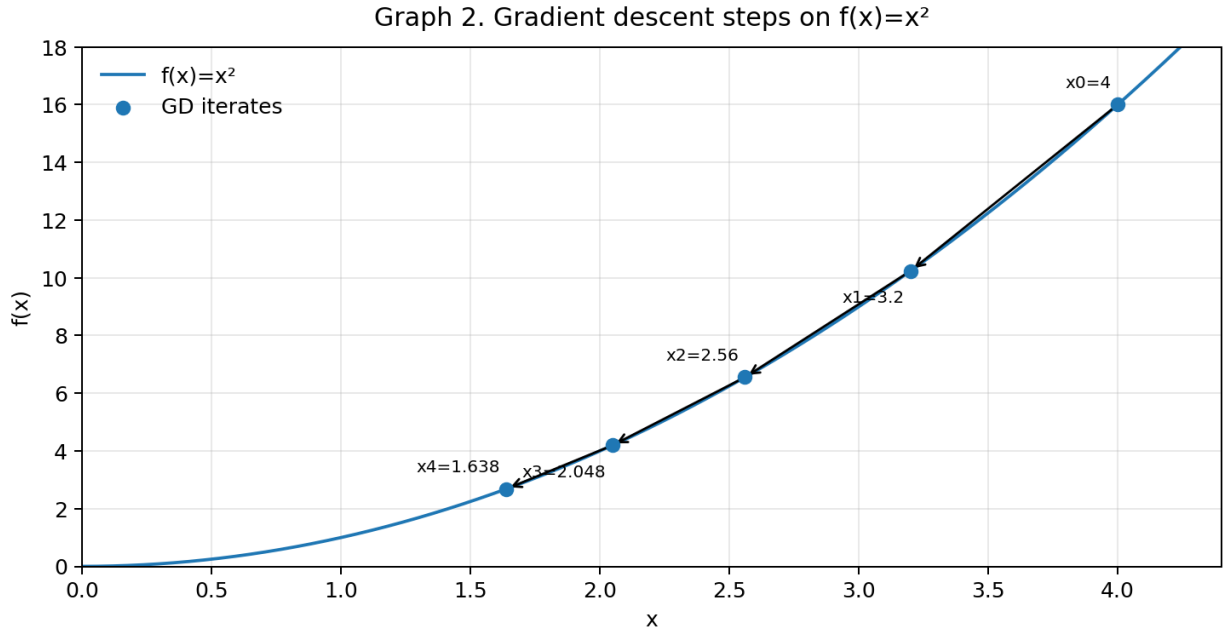


Check your graph: The curve is symmetric around x=0 and has its minimum f(0)=0.

Graph Plotting Exercise 2. Gradient descent on $f(x)=x^2$

Task: Plot the curve $f(x)=x^2$ and mark the gradient-descent iterates for $x_0=4$, $\eta=0.1$.

Suggested points / plot instructions: Use $x_0=4$, $x_1=3.2$, $x_2=2.56$, $x_3=2.048$, $x_4=1.6384$. Plot each (x_t, x_t^2) and draw arrows toward the next iterate.

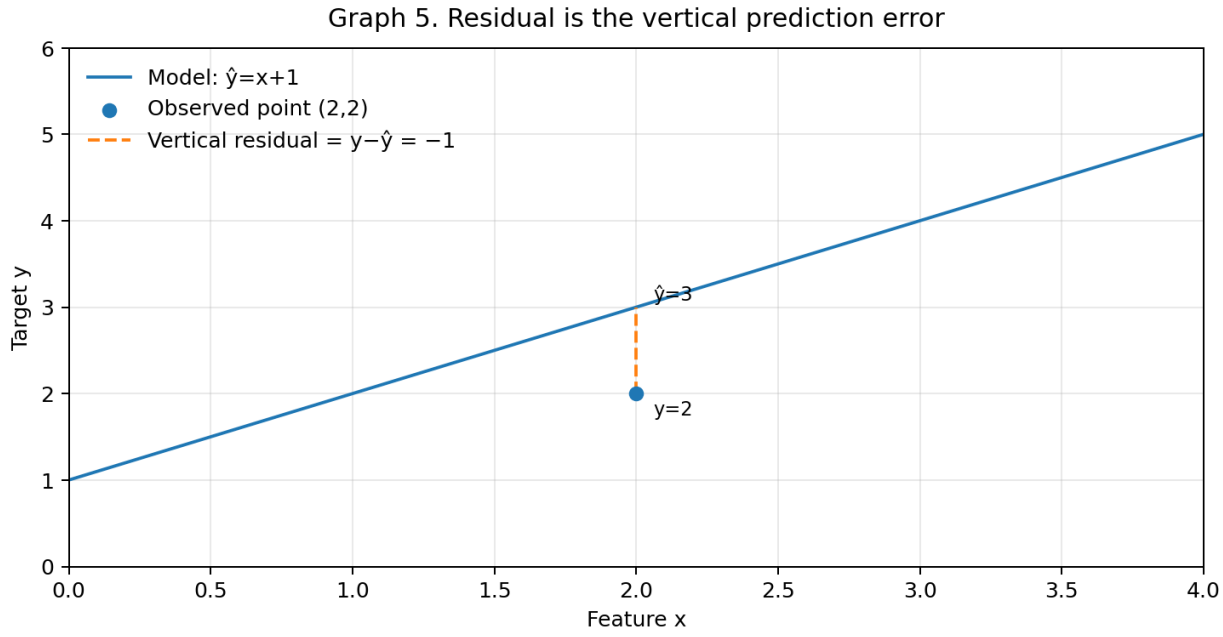


Check your graph: The iterates move toward x=0, where the function is minimized; the objective values decrease.

Graph Plotting Exercise 3. Residual as a vertical distance

Task: Plot the model $\hat{y}=x+1$ and mark observed point (2,2). Show the residual with a dashed vertical line.

Suggested points / plot instructions: At x=2, model prediction is $\hat{y}=3$. Mark observed y=2 and predicted point (2,3), then connect them vertically.

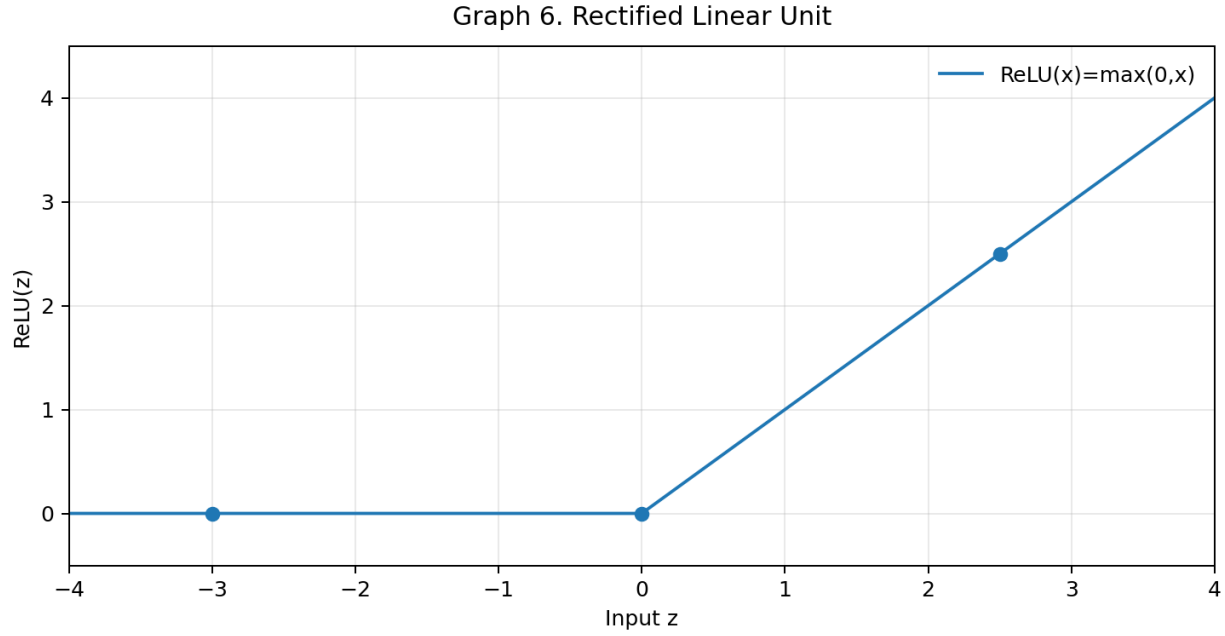


Check your graph: Residual $r=y-\hat{y}=2-3=-1$, so the observed value is one unit below the prediction.

Graph Plotting Exercise 4. ReLU activation

Task: Plot ReLU(z)=max(0,z) for z from −4 to 4.

Suggested points / plot instructions: For negative z, plot y=0. For z=0, y=0. For positive values, use (1,1), (2,2), (3,3).

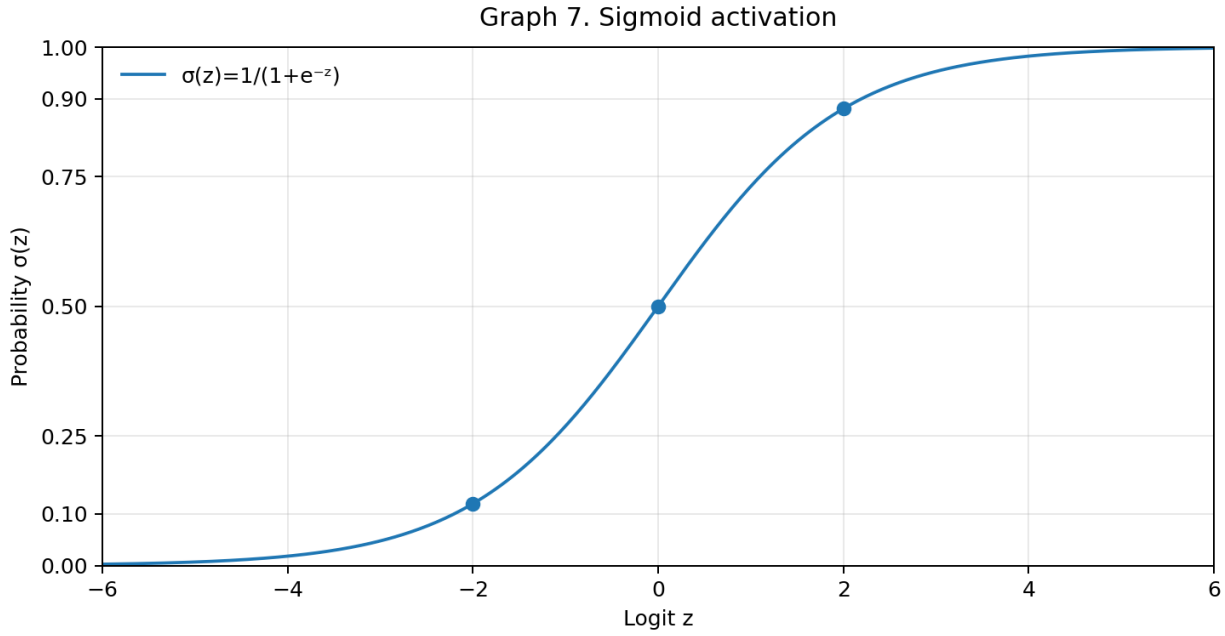


Check your graph: The negative branch is flat at zero; the positive branch follows y=z.

Graph Plotting Exercise 5. Sigmoid activation

Task: Sketch σ(z)=1/(1+e^(−z)) and mark z=−2, 0, 2.

Suggested points / plot instructions: Use approximate points (−2,0.119), (0,0.5), (2,0.881). Draw a smooth S-shaped curve approaching 0 and 1.



Check your graph: The output remains between 0 and 1 and σ(0)=0.5.